INNOVATION SANDBOX INDIVIDUAL ASSIGNMENT

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Section: 3

1. Empathise

Initial problem

The initial problem identified for this project is that patients struggle to keep track of prescriptions, medical reports, appointments and follow-ups, especially when they are managing treatment across multiple healthcare providers. The problem is particularly relevant for people managing chronic or recurring healthcare needs, elderly family members and family members who coordinate healthcare for someone else. This problem is faced by my parents. For primary research, I interviewed them and two other close relatives.

What users experience

Survey responses show that patients may have healthcare information spread across physical files, phone galleries, WhatsApp messages, e-mail, hospital portals, calendars and handwritten notes. Finding an old prescription or remembering a follow-up can require checking several places.

How often it occurs

From the survey responses received, most people reported accessing such information several times a week or a few times a month. The problem can recur whenever a person visits multiple doctors undergoes diagnostic tests, takes medicines for several weeks or months, attends follow-ups, changes prescriptions, or manages healthcare for another family member.

What it disrupts

The fragmented information can increase the time and effort required to retrieve records, remember appointments, and coordinate care.

Why it matters

Healthcare information is often needed at the right time. Difficulty finding or remembering it can create additional stress and effort for patients and family caregivers. The survey indicates that respondents see value in an app that stores healthcare documents, tracks appointments, provides reminders and organized family healthcare information. Many people have provided a rating of 4 and 5 for the need of an app.

What users currently do

From the survey, I found that respondents use physical files/folders, phone galleries, WhatsApp, email/cloud storage and hospital/healthcare apps to manage healthcare information.

What remains unresolved

These alternatives can work independently but do not necessarily provide one simple view of the patient's healthcare journey. This indicates an opportunity for a solution that brings documents, appointments, reminders and family healthcare information together.

2. Define

Problem statement

Patients and family caregivers managing ongoing healthcare struggle to keep prescriptions, reports, appointments and follow-ups organized because information is spread across physical files, phone galleries, messaging apps and different healthcare providers. This creates additional effort and increases the possibility of missing or forgetting important information.

Who experiences the problem?

Adults who manage recurring healthcare needs for themselves or an immediate family member who are tech-savvy.

When or where does the problem occur?

- Visiting multiple doctors
- Undergoing diagnostic tests
- Taking medicines for several weeks or months
- Attending follow up appointments
- Changing prescriptions
- Managing healthcare for parents or other family members
- Trying to retrieve an older medical report

Why does it matter

The user must repeatedly search for information and coordinate several healthcare events across different sources.

Underlying Unmet Need

“I want one simple place that tells me what I need to remember and allows me to quickly find the relevant healthcare information when I need it.”

3. Size the Opportunity

India provides substantial potential user base. The World Bank estimates India's 2025 population at approximately 1.46 billion with around 70% of the population using the Internet. Source: World Bank Open Data.

The potential need is also visible from India's NCD programme. As of June 2026, the Ministry of Health's national NCD portal reported approximately 76.94 crore beneficiaries enrolled and 9.36 crore patients under treatment for hypertension and diabetes. Source: National NCD Programme, Ministry of Health and Family Welfare.

These figures should not be interpreted as the exact market for Caretrack. Instead, they demonstrate that India has a very large population interacting with healthcare systems and managing ongoing treatment. India's digital health ecosystem is also developing. Ayushman Bharat Digital Mission's Personal Health Record framework supports viewing longitudinal records, treatment details, and discharge summaries.

Initial TAM-SAM-SOM Approach

Market level	Definition	Indicative size
TAM	India's internet-using population	~1.02 billion
SAM	Digitally comfortable users who regularly manage healthcare information for themselves or family members	Assumption-based subset of TAM
SOM	Users initially reachable through urban healthcare networks, diagnostic centres, pharmacies, corporate wellness programmes and digital marketing	Initial target: 1 lakh paying users

Initial Revenue Scenario

1 lakh paying users at Rs. 499 per year. Annual revenue potential under this scenario would be Rs. 4.99 crores. This is a scenario-based business assumption, not a prediction, and should be revised after validation.

Users Most Likely to Adopt or Pay

- Adult children managing healthcare for elderly parents

- Chronic disease patients
- Patients undergoing long-term treatment
- Patients consulting multiple specialists
- Families managing healthcare for children
- Elderly patients supported by digitally capable family members
- People undergoing frequent diagnostic testing
- Corporate employees managing healthcare for their families

Among these segments, family caregivers managing parents' healthcare are the initial segment to test because they may handle prescriptions, reports, appointments and follow-ups across multiple healthcare providers.

Opportunity Assumption

The key assumption is not simply that people need healthcare information. ABDM already provides infrastructure for managing health records. The opportunity for CareTrack is that users may still need a simpler personal organization layer that helps them remember upcoming actions and quickly retrieve information when required. This distinction is important, but CareTrack is positioned as a complement to existing digital health infrastructure rather than another medical record system.

4. Ideate

Synthesize Market Information

The market information suggests that CareTech should focus on the organization and retrieval of healthcare information rather than attempting to pick up a medical diagnosis or treatment platform. India's large digitally connected population, growing digital-health infrastructure and significant population managing recurring healthcare needs create a relevant context for the solution.

Existing digital health infrastructure such as ABDM addresses access to digital health records, while the identified user problem is more specifically about remembering appointments, managing prescriptions, finding reports, and coordinating healthcare information across family members.

User segments and Contexts

Segment	Main situation	Potential need
Chronic-care patient	Long-term medication and follow-ups	Reminders + timeline
Elderly patient	Multiple medicines and appointments	Simple reminders
Adult child managing	Information spread across	Family profile + document

parents	family members	storage
Multi-specialist patient	Different doctors and prescriptions	Centralized timeline
Parents managing children's healthcare	Reports and medical records	Family profiles
Frequent diagnostic-test user	Regular blood tests/scans	Report retrieval
Corporate employee	Managing own and family healthcare	One consolidated view
Long-term treatment patient	Healthcare information accumulated over months	Search + timeline

Alternative Solutions

Idea	**Description**
Digital Health Locker	A single place to upload prescriptions, reports and medical documents.
Medication Reminder	Focus specifically on medicine schedules and refill reminders.
Healthcare Timeline	Automatically organise appointments, prescriptions, reports and treatment events chronologically.
Family Healthcare Dashboard	Allow one user to manage healthcare information for parents, children and themselves.
AI Healthcare Assistant	Allow users to upload documents and ask questions such as "What was my previous prescription?" or "When is my next follow-up?"
CareTrack	Combine document storage, medication reminders, appointments and a simple healthcare timeline.

Challenge Assumptions

Assumption 1: More features automatically create more value. A broader product may become difficult to use, so the MVP should focus on the most frequent organizational problem.

Assumption 2: Users need another medical record system. Existing digital health infrastructure means Caretrack should not try to replace official health records. Its role should be a simple personal and family organization layer.

Assumption 3: AI should be included from the beginning. Although an AI assistant could make the product more powerful, medical interpretation creates additional equity, privacy, and safety concerns. Therefore, AI functionality was not prioritized for the first MVP.

Assumption 4: Everyone managing Healthcare is equally likely to pay. Occasional healthcare users may have limited need for a dedicated tool. Family caregivers and people managing recurring treatment may have a stronger reason to adopt it.

Assumption 5: The initial Rs. 4.99 annual price is validated. This is only a business assumption and must be tested through validation interviews. Interest in the product does not necessarily mean willingness to pay.

Identify Risks and Barriers

The main risks and barriers identified during ideation were:

Risk / Barrier	**Potential Impact**	**Initial Response**
Privacy concerns	Users may hesitate to store healthcare information	Consent, encryption and minimum data collection
Existing alternatives	Users may continue using WhatsApp, galleries and physical files	Focus on easier organization and retrieval
ABDM overlap	Users may see CareTrack as unnecessary	Position it as a complementary organization layer
Low willingness to pay	Users may prefer free solutions	Test freemium and premium pricing
Low adoption	Users may not want another healthcare app	Start with high-need segments
Medical safety	AI interpretation could create harmful errors	Exclude diagnosis and treatment recommendations
Data-entry effort	Uploading information may feel inconvenient	Keep upload and search simple

Evaluation Matrix

Solution	Feasibility	Cost	User impact	Adoption	Scalability	Risk	Time
Digital Health Locker	5	5	3	4	5	4	5
Medication Reminder	5	5	4	5	5	3	5
Healthcare Timeline	4	4	5	4	5	4	4
Family Dashboard	4	4	5	4	5	4	4
AI Healthcare Assistant	3	3	5	3	5	2	2
CareTrack MVP	5	4	5	4	5	4	4

Each criterion was scored from 1 (low) to 5 (high), based on expected MVP development requirements and user value.

Selected Solution

The selected concept is CareTrack, a lightweight healthcare management assistant combining the most useful features without attempting to become a doctor or diagnostic tool. CareTrack was selected because it combines several high-frequency organizational needs without trying to make medical decisions. It brings together document storage, medication reminders, appointments, family profiles, and a healthcare timeline.

Rejected Idea

The AI healthcare assistant was not selected as the initial MVP. Although conversational AI could make the product more powerful, it introduces greater safety, accuracy, privacy, and regulatory concerns. The first version should solve the simpler organizational problem before introducing AI based interpretation.

5. Prototype

Prototype Type: Process Flow

What is CareTrack?

CareTrack is a simple mobile healthcare management assistant that helps users organize prescriptions, medical reports, appointments, medication reminders, follow up dates, and family members healthcare information.

Core user journey

Upload – Organize – Remind – Retrieve

Example: User visits doctor → uploads prescription → enters medicines → adds follow-up date → receives reminder → later searches "Dermatology" → sees prescription, reports and appointment history.

Who is CareTrack for?

CareTrack is primarily designed for people who manage recurring healthcare needs for themselves or their family members.

MVP Features

Feature	Purpose
Health Timeline	Shows important healthcare events chronologically.
Document Vault	Stores prescriptions, reports and discharge summaries.
Medication Reminder	Provides simple reminders based on the user's entered prescription.
Appointment Tracker	Stores doctor, date, time and purpose.
Family Profiles	Allows separate records for parents, spouse, children or other family members.
Search	Searches by doctor, date, document type or condition.

MVP Exclusions

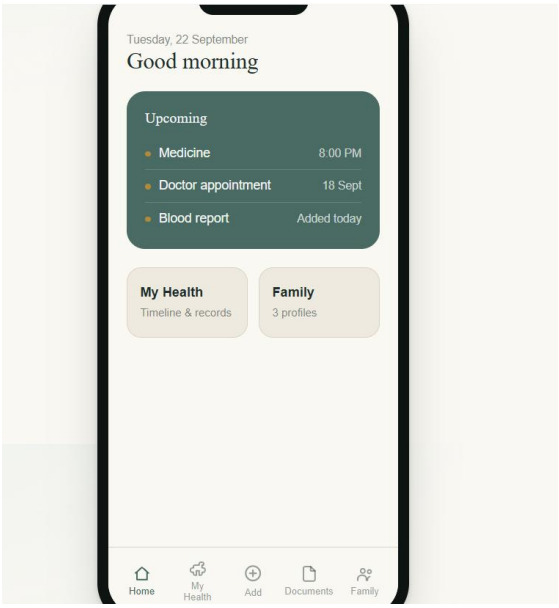
- Diagnosing diseases
- Recommending medicines
- Changing prescriptions
- Interpreting medical conditions independently
- Replacing doctors
- Making emergency medical decisions

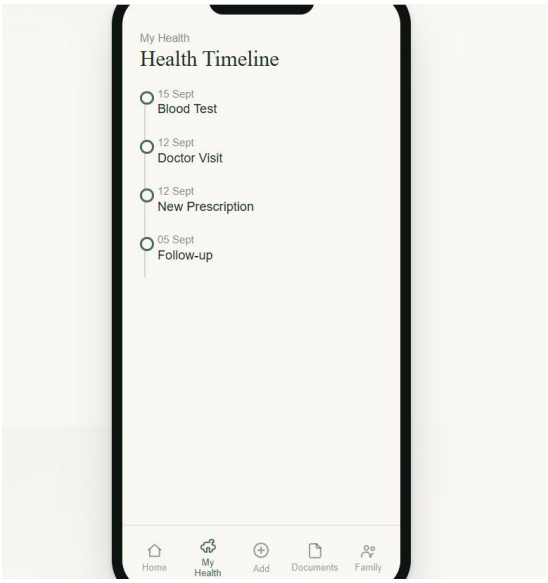
- This keeps the initial product focused and reduces potential safety risks

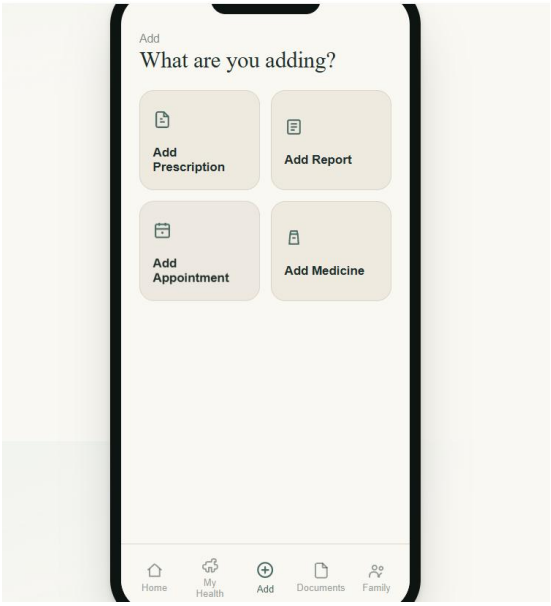
Prototype Process Flow

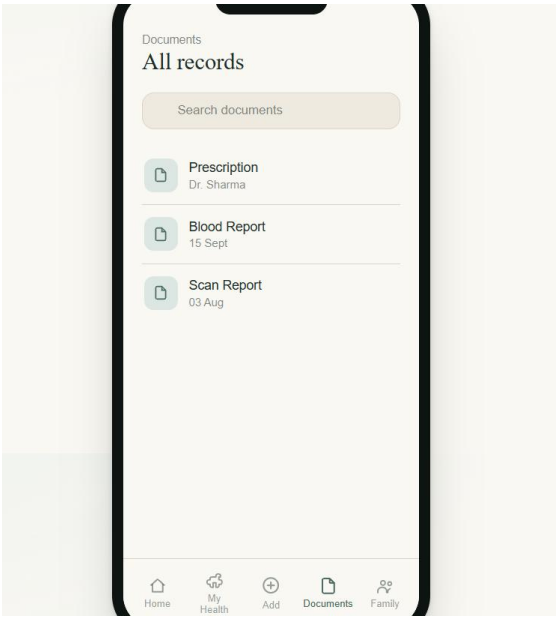
Doctor Visit
↓
Upload Prescription
↓
Enter Medication Details
↓
Add Follow-up Date
↓
Receive Reminder
↓
Search Healthcare Information
↓
Retrieve Prescription / Report / Appointment History

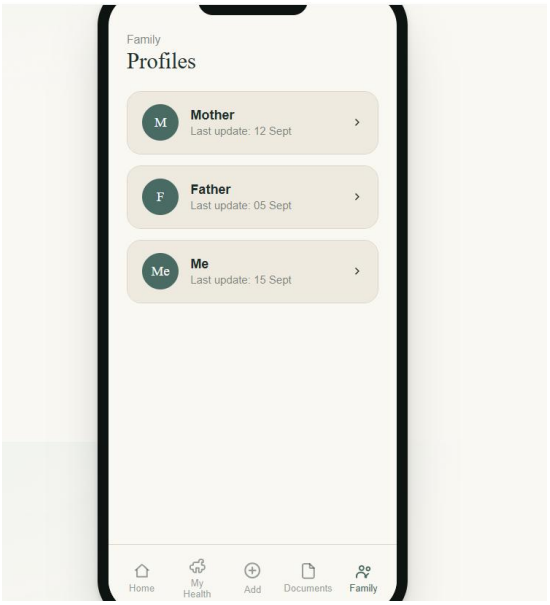
Prototype screenshots




My Health
Health Timeline
15 Sept
Blood Test
12 Sept
Doctor Visit
12 Sept
New Prescription
05 Sept
Follow-up
Home
My Health
Add
Documents
Family


Add
What are you adding?
Add Prescription
Add Report
Add Appointment
Add Medicine
Home
My Health
Add
Documents
Family


Documents
All records
Search documents
Prescription
Dr. Sharma
Blood Report
15 Sept
Scan Report
03 Aug
Home
My Health
Add
Documents
Family



MVP Cost

Component	Approx. cost
Figma/Canva prototype	₹0–₹2,000
No-code database	₹0–₹5,000
Basic interface development	₹10,000–₹25,000
User testing	₹0–₹5,000
Estimated prototype/MVP range	₹10,000–₹37,000

Basis for the estimate

The estimate assumes that the first version would be developed using low code/no code tools, a basic database, a simple interface and a limited user testing. It is intended for an academic MVP rather than a commercially launched healthcare platform. The estimate excludes the cost of a full development team, large scale infrastructure and regulatory work. These figures are planning assumptions rather than vendor quotations and would need to be validated before commercial development.

Initial users or audience it could reach

For an initial pilot, CareTrack could target 50 to 100 users who regularly manage healthcare information for themselves or family members. The pilot would mainly focus on adult children managing elderly parents, people undergoing recurring

treatment and people who regularly manage prescriptions and diagnostic reports. Potential channels for reaching these users could include personal networks, healthcare communities, diagnostic centers, pharmacies, hospitals and digital healthcare communities. The purpose of the initial pilot would be to test whether users experience the identified problem, understand the proposed workflow and are willing to use the solution.

6. Test and Validate

I rolled out the survey to more people and got a total of 17 responses. Google form responses link: https://docs.google.com/spreadsheets/d/1yBt6H64liVRnBVRXxUEJ-Wlsc6Md27v1KWcraRLlkiI/edit?usp=sharing

Below are the findings from the survey:

What was confirmed

The research supported the assumption that some users experience difficulty managing and retrieving healthcare information. Several respondents reported moderate, high difficulty and recurring problems with their current methods.

What was challenged

The assumption that the healthcare information management is a frequent problem for everyone was challenged. Some respondents accessed healthcare information only monthly or rarely. This suggests that Caretrack may be more relevant to users who manage healthcare information frequently or for multiple family members.

What surprised me

Respondents used a mixture of physical and digital methods, including physical files, WhatsApp, phone galleries, e-mail, cloud storage and healthcare apps. Privacy and security also emerged as a consistent concern of using an app to manage healthcare documents.

What I changed after receiving feedback

Based on the feedback, I would place greater emphasis on privacy and security in Caretrack. I would also focus the initial target audience on people who manage healthcare information for multiple family members or access it regularly. The solution would continue to focus on organizing, storing, retrieving and reminding users about healthcare information rather than providing medical diagnosis or treatment recommendations.

7. Innovation Story

The Problem

Healthcare generates a large amount of information, but patients often manage it across multiple disconnected places. Prescriptions may be in physical files, reports in phone galleries, appointments in calendars and instructions in WhatsApp messages.

Human Insight

The problem is not necessarily a lack of healthcare information. It is the difficulty of managing information at the right time.

Opportunity

India's large digitally connected population and expanding digital health infrastructure create a potential environment for a consumer healthcare management tool. The World Bank estimates India's 2025 population at approximately 1.46 billion with Internet usage around 70%. The government's NCD program also reports large numbers of people being managed for chronic conditions.

Alternatives Explored

- Document storage
- Medication reminders
- Healthcare timelines
- Family dashboards
- AI healthcare assistants

The final concept combines the most useful organizational functions while deliberately avoiding medical diagnosis or treatment decisions.

Proposed solution

CareTrack is a simple healthcare management assistant that helps users store, organize, and retrieve healthcare information while reminding them about important events.

User Transformation

Before	After
"Where did I keep that prescription?" "When is my next appointment?" "Which report did the doctor ask for?" "I need to send my mother's previous reports."	Open CareTrack → select family member → view timeline → find prescription/report → check appointment/reminder.

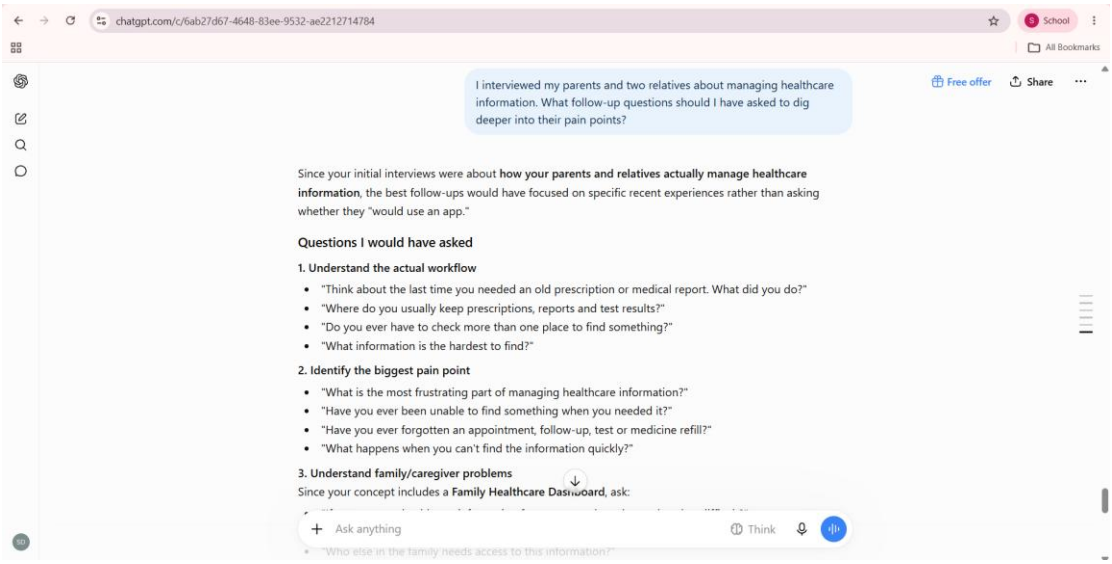
Value proposition: "Your family's healthcare information, organised in one simple place."

AI Tools used

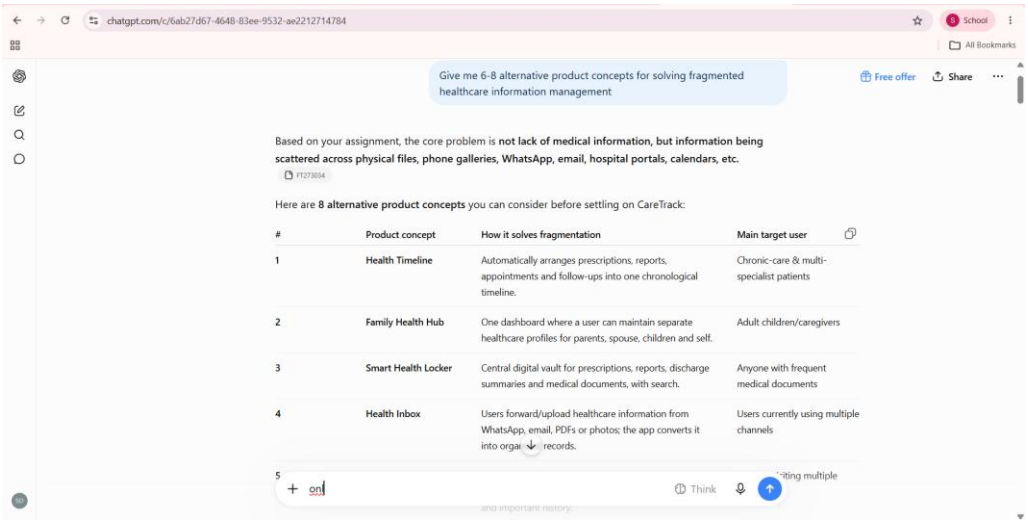
I used ChatGPT for brainstorming ideas, analysis and refining the content of the assignment.

AI Conversation Log

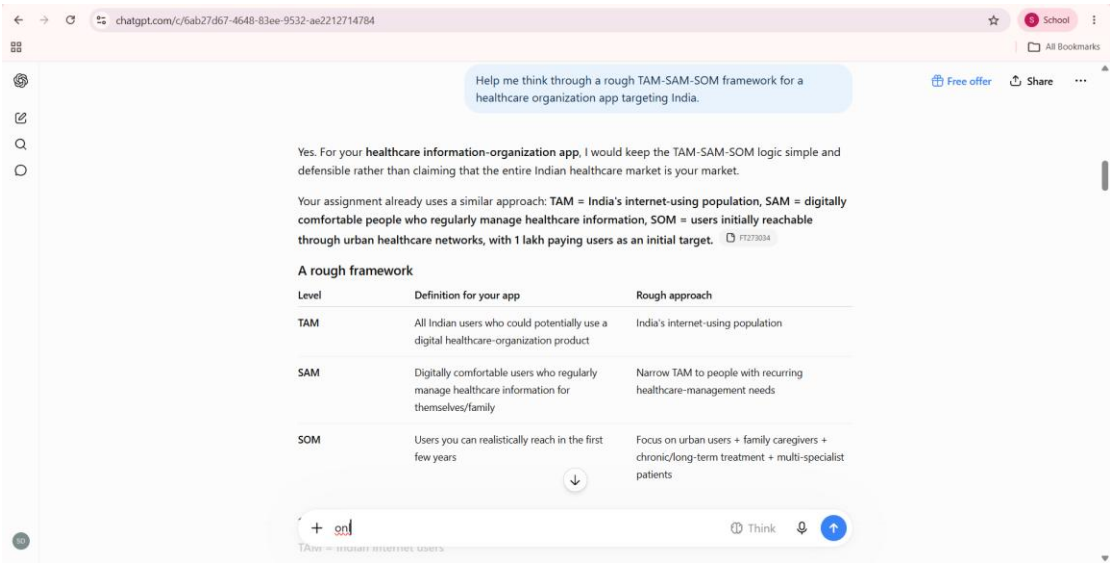
Prompt: I interviewed my parents and two relatives about managing healthcare information. What follow-up questions should I have asked to dig deeper into their pain points?



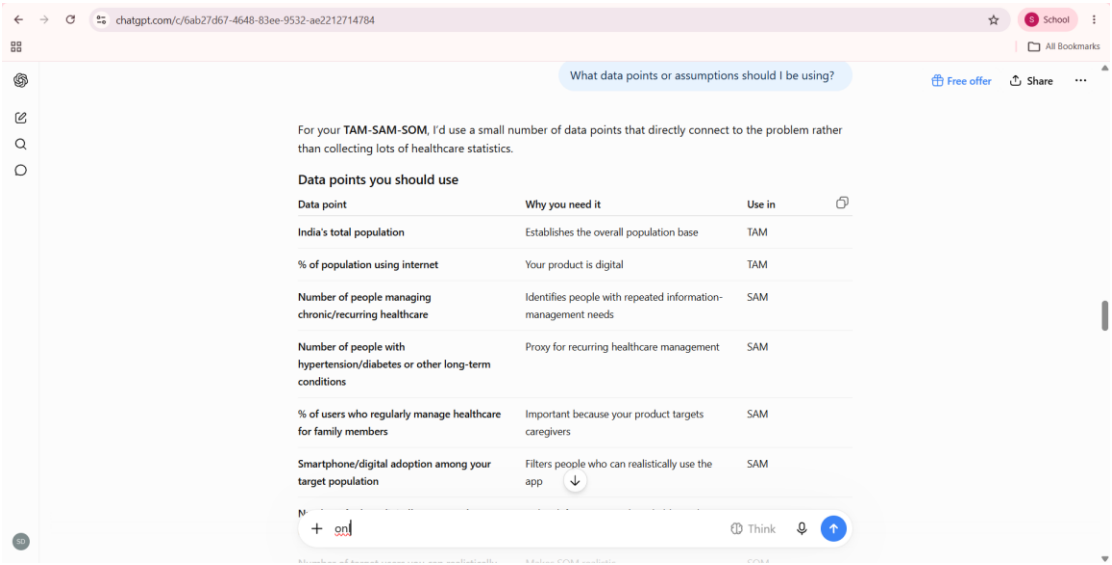
Prompt: Give me 6-8 alternative product concepts for solving fragmented healthcare information management



Prompt: Help me think through a rough TAM-SAM-SOM framework for a healthcare organization app targeting India.



Prompt: What data points or assumptions should I be using?



Prompt: I have six potential solutions for a healthcare organization app:

Digital Health Locker

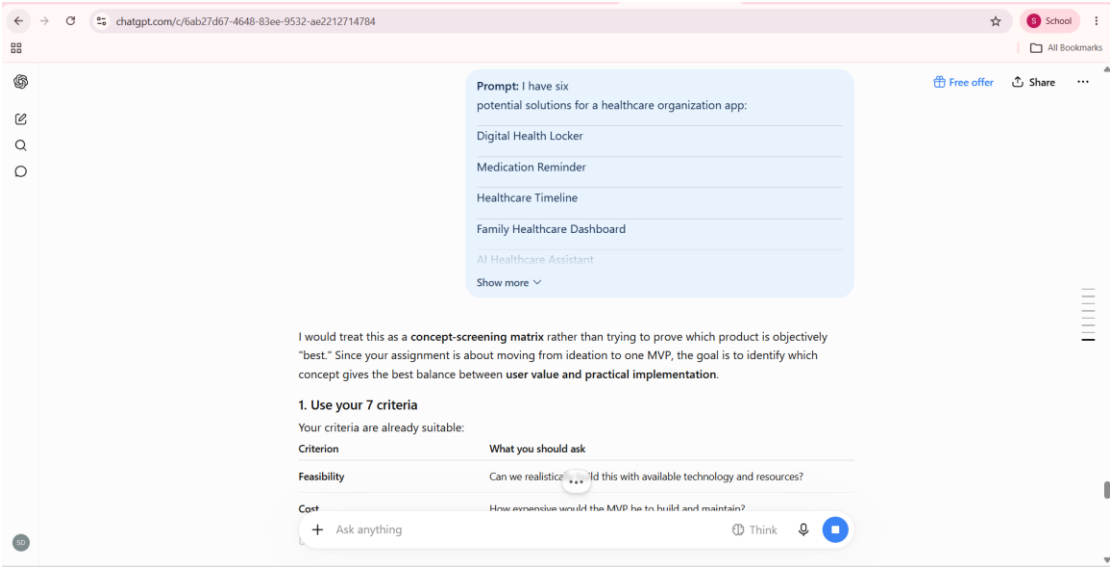
Medication Reminder

Healthcare Timeline

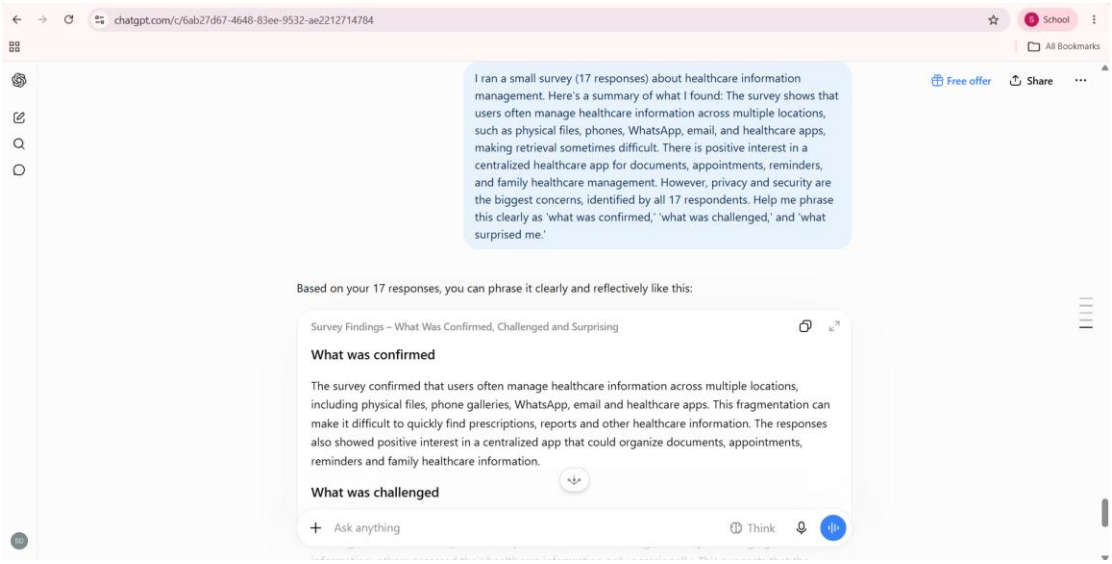
Family Healthcare Dashboard

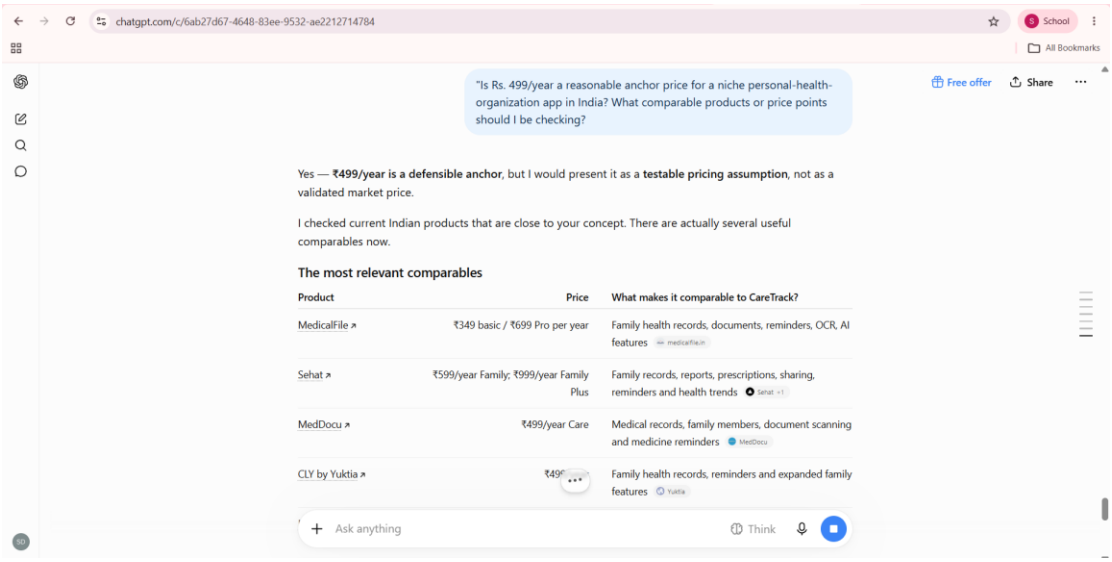
AI Healthcare Assistant

CareTrack . I want to evaluate them on - feasibility, cost, user impact, adoption, scalability, risk, time? How should I go about it?



Prompt: I ran a small survey (17 responses) about healthcare information management. Here's a summary of what I found: The survey shows that users often manage healthcare information across multiple locations, such as physical files, phones, WhatsApp, email, and healthcare apps, making retrieval sometimes difficult. There is positive interest in a centralized healthcare app for documents, appointments, reminders, and family healthcare management. However, privacy and security are the biggest concerns, identified by all 17 respondents. Help me phrase this clearly as 'what was confirmed,' 'what was challenged,' and 'what surprised me.'





Prompt: Is Rs. 499/year a reasonable anchor price for a niche personal-health-organization app in India? What comparable products or price points should I be checking?